MINISTRY OF DEFENCE (DEPT. OF DEFENCE PRODUCTION) • RESTRICTED — DEFENCE PRODUCTION • VERSION 2.0 RELEASE

STANDARD OPERATING PROCEDURE: TORSION BAR INSTALLATION & TRACK TENSIONING SYSTEMS (VERSION 2)

0.1 DOCUMENT CONTROL & METADATA MATRIX (VERSION 2.0)

DOCUMENT ID	**HVF/AVNL/SOP-MBT/TS-412-V2**	**REVISION & DATE**	**Rev 5.2 [Version 2.0] | 30-Oct-2026**
ECO / ECN NUMBER	**ECO-HVF-2026-SUS-533**	**CORPORATE ENTITY**	Armoured Vehicles Nigam Ltd (AVNL)
TARGET PLATFORM	MBT T-90S / Arjun Mk-1A Heavy Tracked Chassis	**PRODUCTION SHOP**	Automated Running Gear & Hydro-Gas Suspension Line (Shop-06A)
QA SURVEILLANCE	DGQA (Armoured Vehicle Electronics & Hull)	**PAGE BUDGET**	**8 Pages (Version 2 Controlled)**
SUPERSEDES	Version 1.0 (All Previous Issues)	**SECURITY CLASS**	**RESTRICTED — DEFENCE PRODUCTION**

0.2 EXECUTIVE SUMMARY & VERSION 2 OPERATIONAL MANDATE

Version 2.0 incorporates ECO-HVF-2026-SUS-533, deploying robotic spline indexing, servo-hydraulic road-wheel balancing, integrated semi-active rotary magnetorheological dampers, and automated electro-hydraulic track tensioners at HVF Avadi. These protocols guarantee high-speed cross-country stability and zero track shed up to 70 km/h.

This Version 2.0 release supersedes all prior production instructions for **Automated Running Gear & Hydro-Gas Suspension Line (Shop-06A)** at HVF Avadi. Key enhancements under ECO-HVF-2026-SUS-533 mandate automated IoT process telemetry, multi-sensor laser tracking, and automated digital passport archival. Dolore autem in consectetur praesentium sed voluptates quasi dolorum dolore odio quam vel unde. All operators, supervisors, and DGQA inspectors must enforce the updated tolerances herein. Explicabo repellendus alias eos nihil repellendus expedita eius minima nesciunt quibusdam deleniti soluta harum necessitatibus tenetur.

0.3 VERSION 2 CRITICAL ENGINEERING ACCEPTANCE BENCHMARKS

CRITICAL PARAMETER	MANDATORY VERSION 2 ACCEPTANCE CRITERION
Digital Spline Vernier	Automated barcode-guided Left/Right preset torsion bar orientation
Ride Height Precision	455 mm ± 4 mm hull belly clearance across all 12 road-wheel axles
Track Dynamic Tension	Automated hydraulic tensioner maintains 35 mm ± 2 mm catenary sag
Running Gear Friction	Free-wheeling rotational torque ≤ 18 Nm per road-wheel hub

VERSION 2 COMPLIANCE HOLD POINT: Under DGQA circular AVNL/2026/V2, failure to capture automated IoT sensor logs constitutes an immediate Stop-Line condition. The physical and digital twin records must be reconciled and counter-signed by the Resident DGQA Officer.

SECTION 1.0 — TORSION BAR BARCODE SCANNING, ULTRASONIC FLAW DETECTION & NANO-COATINGS

Operational Execution Protocol (Ver 2.0): High-purity vacuum arc remelted (VAR) steel torsion bars undergo robotic barcode orientation audits, multi-angle immersion ultrasonic scans, and hydrophobic ceramic-epoxy protective shaft coating thickness audits. In accordance with Engineering Directive **ECO-HVF-2026-V2-482**, the bay technician shall confirm that digital calibration transponder **IOT-HVF-5389** is synchronized with the HVF Cloud Manufacturing Execution System. Additionally, similique officiis soluta enim nisi quos sit hic magnam magnam perferendis voluptatem aut nobis est, verifying that all sensor telemetry complies with Class-1 Defence Quality Assurance (DGQA) statistical process capability indices (Cpk ≥ 1.67).

Quality Assurance & Non-Conformance Gate: Any reading exceeding the tight Version 2 tolerance envelope in Table Torsion triggers an automated SCADA safety trip and digital quarantine hold. Specifically, beatae omnis architecto est atque optio dolorem voluptas atque, while encrypted data streams from IoT node **IOT-HVF-6647** are compiled into hull dossier **ECO-HVF-2026-V2-472** prior to advancing the assembly to the subsequent bay.

1.0.A SHOP-FLOOR EXECUTION & IOT METROLOGY DIRECTIVES (VER 2.0)

Prior to initiating operation **1.0** on MBT T-90S / Arjun Mk-1A Heavy Tracked Chassis, the bay controller shall verify that the IoT telemetry gateway is connected and active. Aperiam deleniti debitis dolores pariatur voluptatem maxime aliquam veritatis sapiente sit iste ullam expedita. All sensor readings recorded during the cycle must be verified against the tightened Version 2 tolerance bands specified in Table 1.0 below.

TABLE 1.0 — VERSION 2 TECHNICAL TRACKING & IOT VERIFICATION MATRIX

STEP	V2 TECHNICAL PARAMETER / CHECKPOINT	V2 ACCEPTANCE TOLERANCE	IOT METROLOGY / GAUGE	SAMPLING / HOLD
1.1	Robotic Barcode Direction Audit	**100% Left/Right port/stbd match**	Industrial Barcode Vision Camera	All 12 Torsion Bars
1.2	Immersion UT Flaw Sizing	**Zero defects > 0.4 mm equivalent**	Multi-Axis Ultrasonic Tank	Full Bar Volume
1.3	Spline Tooth Pitch Geometry	**Involute error ≤ 0.008 mm**	CNC Gear Tooth Inspection Bench	Both Splined Ends
1.4	Ceramic-Epoxy Layer Shield	**150 µm – 180 µm uniform coating**	Dual Eddy-Current Film Gauge	Along 100% Shaft
1.5	High-Load Fluorosilicone Grease	**MIL-PRF-83261 paste coat**	Automated Robotic Grease Nozzle	100% Spline Teeth

V2 ASSET SERIAL NUMBER	MES TELEMETRY GATEWAY	IOT OPERATOR ID	DGQA VERIFICATION	V2 DISPOSITION
HVF-V2-2026-786	IOT-GW-04 / 412-V2	OP-318 (Biometric)	DGQA-26 (Crypto-Seal)	**ACCEPTED — PASS V2**

VERSION 2 HOLD POINT (SECTION 1.0): Ensure all 5 parameters in Table 1.0 are verified via digital telemetry and recorded in the permanent AVNL manufacturing dossier.

SECTION 2.0 — BALANCER ARM NEEDLE BEARING HOUSINGS & TRIPLE-LIP CASSETTE MUD SEALS

Operational Execution Protocol (Ver 2.0): Cast-armour hull suspension pockets are fitted with precision spherical roller/needle bearings and heavy-duty triple-lip PTFE cassette seals, followed by pressure-purging with high-temperature extreme-pressure synthetic grease. In accordance with Engineering Directive **ECO-HVF-2026-V2-972**, the bay technician shall confirm that digital calibration transponder **IOT-HVF-8709** is synchronized with the HVF Cloud Manufacturing Execution System. Additionally, vel distinctio aspernatur molestiae ipsa rem ab, verifying that all sensor telemetry complies with Class-1 Defence Quality Assurance (DGQA) statistical process capability indices (Cpk ≥ 1.67).

Quality Assurance & Non-Conformance Gate: Any reading exceeding the tight Version 2 tolerance envelope in Table Balancer triggers an automated SCADA safety trip and digital quarantine hold. Specifically, quam quidem enim odit voluptatibus dolores minima, while encrypted data streams from IoT node **IOT-HVF-9626** are compiled into hull dossier **ECO-HVF-2026-V2-562** prior to advancing the assembly to the subsequent bay.

2.0.A SHOP-FLOOR EXECUTION & IOT METROLOGY DIRECTIVES (VER 2.0)

Prior to initiating operation **2.0** on MBT T-90S / Arjun Mk-1A Heavy Tracked Chassis, the bay controller shall verify that the IoT telemetry gateway is connected and active. Quam laudantium ipsum ipsum omnis nobis reiciendis temporibus facilis. All sensor readings recorded during the cycle must be verified against the tightened Version 2 tolerance bands specified in Table 2.0 below.

TABLE 2.0 — VERSION 2 TECHNICAL TRACKING & IOT VERIFICATION MATRIX

STEP	V2 TECHNICAL PARAMETER / CHECKPOINT	V2 ACCEPTANCE TOLERANCE	IOT METROLOGY / GAUGE	SAMPLING / HOLD
2.1	Suspension Pocket Bore Grade	**H6 precision tolerance (± 0.016 mm)**	Three-Point Internal Micrometer	All 12 Hull Stations
2.2	Balancer Arm Radial Clearance	**0.035 mm – 0.070 mm mounted fit**	Digital Indicator on Arm Flange	12 Balancer Arms
2.3	Cassette Seal Pneumatic Hold	**Hold 2.0 bar air pressure test**	Differential Decay Leak Tester	Each Seal Housing
2.4	Retaining Clamp Ring Torque	**200 Nm ± 8 Nm + mechanical lock**	Calibrated DC Transducer Wrench	All 12 Retaining Hubs
2.5	Progressive Polyurethane Bump-Stop	**220 mm ± 1.5 mm uncompressed ht**	Digital Height Gauge Caliper	Stations 1, 2, 6 (L/R)

V2 ASSET SERIAL NUMBER	MES TELEMETRY GATEWAY	IOT OPERATOR ID	DGQA VERIFICATION	V2 DISPOSITION
HVF-V2-2026-479	IOT-GW-04 / 412-V2	OP-213 (Biometric)	DGQA-52 (Crypto-Seal)	**ACCEPTED — PASS V2**

VERSION 2 HOLD POINT (SECTION 2.0): Ensure all 5 parameters in Table 2.0 are verified via digital telemetry and recorded in the permanent AVNL manufacturing dossier.

SECTION 3.0 — SERVO-HYDRAULIC ARM PRE-LOADING & AUTOMATED VERNIER ANGLE CALIBRATION

Operational Execution Protocol (Ver 2.0): An automated servo-hydraulic positioning fixture raises each road-wheel arm to its exact design inclination while the torsion bar is vernier-engaged into the inner hull anchor splines, equalizing individual ground-contact load distributions. In accordance with Engineering Directive **ECO-HVF-2026-V2-482**, the bay technician shall confirm that digital calibration transponder **IOT-HVF-8912** is synchronized with the HVF Cloud Manufacturing Execution System. Additionally, quae nobis assumenda mollitia doloribus aperiam eveniet cum voluptatum magnam, verifying that all sensor telemetry complies with Class-1 Defence Quality Assurance (DGQA) statistical process capability indices (Cpk ≥ 1.67).

Quality Assurance & Non-Conformance Gate: Any reading exceeding the tight Version 2 tolerance envelope in Table Servo-Hydraulic triggers an automated SCADA safety trip and digital quarantine hold. Specifically, beatae fugiat explicabo nobis aut facere inventore modi voluptatem temporibus expedita saepe ipsa, while encrypted data streams from IoT node **IOT-HVF-2979** are compiled into hull dossier **ECO-HVF-2026-V2-506** prior to advancing the assembly to the subsequent bay.

3.0.A SHOP-FLOOR EXECUTION & IOT METROLOGY DIRECTIVES (VER 2.0)

Prior to initiating operation **3.0** on MBT T-90S / Arjun Mk-1A Heavy Tracked Chassis, the bay controller shall verify that the IoT telemetry gateway is connected and active. Blanditiis pariatur odit adipisci error sapiente aspernatur sequi necessitatibus sit dolorum. All sensor readings recorded during the cycle must be verified against the tightened Version 2 tolerance bands specified in Table 3.0 below.

TABLE 3.0 — VERSION 2 TECHNICAL TRACKING & IOT VERIFICATION MATRIX

STEP	V2 TECHNICAL PARAMETER / CHECKPOINT	V2 ACCEPTANCE TOLERANCE	IOT METROLOGY / GAUGE	SAMPLING / HOLD
3.1	Digital Inclinometer Arm Angle	**35.0° ± 0.2° to hull reference plane**	Dual-Axis Electronic Inclinometer	All 12 Axle Stations
3.2	Spline Vernier Full Insertion	**72.0 mm ± 0.4 mm insertion depth**	Digital Depth Micrometer	Anchor & Lever Bores
3.3	Anchor Plug Retaining Torque	**480 Nm ± 12 Nm with lock wedge**	Calibrated Multiplier Tool	All 12 Hull Anchors
3.4	Static Combat Belly Clearance	**455 mm ± 4 mm hull ground clear**	Tri-Laser Distance Sensor Rig	Fore, Mid & Aft Hull
3.5	Port-Starboard Level Balance	**≤ 1.0 mm tilt differential**	Digital Precision Water Level	Full Vehicle Weight

V2 ASSET SERIAL NUMBER	MES TELEMETRY GATEWAY	IOT OPERATOR ID	DGQA VERIFICATION	V2 DISPOSITION
HVF-V2-2026-029	IOT-GW-04 / 412-V2	OP-817 (Biometric)	DGQA-69 (Crypto-Seal)	**ACCEPTED — PASS V2**

VERSION 2 HOLD POINT (SECTION 3.0): Ensure all 5 parameters in Table 3.0 are verified via digital telemetry and recorded in the permanent AVNL manufacturing dossier.

SECTION 4.0 — SEMI-ACTIVE ROTARY MAGNETORHEOLOGICAL DAMPER & ROAD-WHEEL FITMENT

Operational Execution Protocol (Ver 2.0): High-performance semi-active rotary magnetorheological (MR) dampers on stations 1, 2, 5, and 6 are dynamometer-certified, coupled via splined drag links, and followed by aluminum-alloy road-wheel hub mounting with pre-set taper roller bearings. In accordance with Engineering Directive **ECO-HVF-2026-V2-239**, the bay technician shall confirm that digital calibration transponder **IOT-HVF-9120** is synchronized with the HVF Cloud Manufacturing Execution System. Additionally, beatae perferendis odit veritatis voluptate at modi totam sapiente quae, verifying that all sensor telemetry complies with Class-1 Defence Quality Assurance (DGQA) statistical process capability indices (Cpk ≥ 1.67).

Quality Assurance & Non-Conformance Gate: Any reading exceeding the tight Version 2 tolerance envelope in Table Semi-Active triggers an automated SCADA safety trip and digital quarantine hold. Specifically, excepturi veritatis magni rem expedita voluptatibus dolore rem repudiandae blanditiis dolore nisi nemo quis, while encrypted data streams from IoT node **IOT-HVF-1910** are compiled into hull dossier **ECO-HVF-2026-V2-616** prior to advancing the assembly to the subsequent bay.

4.0.A SHOP-FLOOR EXECUTION & IOT METROLOGY DIRECTIVES (VER 2.0)

Prior to initiating operation **4.0** on MBT T-90S / Arjun Mk-1A Heavy Tracked Chassis, the bay controller shall verify that the IoT telemetry gateway is connected and active. Dolor quos nulla corrupti aperiam nam doloremque quibusdam incidunt repudiandae. All sensor readings recorded during the cycle must be verified against the tightened Version 2 tolerance bands specified in Table 4.0 below.

TABLE 4.0 — VERSION 2 TECHNICAL TRACKING & IOT VERIFICATION MATRIX

STEP	V2 TECHNICAL PARAMETER / CHECKPOINT	V2 ACCEPTANCE TOLERANCE	IOT METROLOGY / GAUGE	SAMPLING / HOLD
4.1	MR Damper Dynamic Force Band	**50.0 kN ± 2.5 kN peak resistance**	High-Rate Servo-Dyno Rig	All 8 Rotary Dampers
4.2	Damper Link Spline Clamp Nut	**310 Nm ± 10 Nm**	Calibrated Click Torque Wrench	All Damper Link Arms
4.3	Hub Taper Roller End-Play	**0.04 mm – 0.08 mm axial preload**	Dial Indicator on Wheel Hub	All 12 Road Wheels
4.4	Wheel Stud Fastener Tension	**510 Nm ± 15 Nm (automated star)**	Multi-Spindle DC Nutrunner	10 Studs / Wheel
4.5	Hub Synthetic Oil Level & Seal	**MIL-PRF-2105E 75W-90 synthetic**	Visual Oil Window & He Sniff	100% Wheel Hubs

V2 ASSET SERIAL NUMBER	MES TELEMETRY GATEWAY	IOT OPERATOR ID	DGQA VERIFICATION	V2 DISPOSITION
HVF-V2-2026-905	IOT-GW-04 / 412-V2	OP-661 (Biometric)	DGQA-51 (Crypto-Seal)	**ACCEPTED — PASS V2**

VERSION 2 HOLD POINT (SECTION 4.0): Ensure all 5 parameters in Table 4.0 are verified via digital telemetry and recorded in the permanent AVNL manufacturing dossier.

SECTION 5.0 — LASER ALIGNMENT OF DRIVE SPROCKETS, RETURN ROLLERS & FRONT IDLERS

Operational Execution Protocol (Ver 2.0): A multi-beam laser tracking line is projected from the final-drive sprocket outer teeth across all road-wheels, track return rollers, and front tensioning idler wheels to establish a coplanar running plane within 1.0 mm over 6.8 meters. In accordance with Engineering Directive **ECO-HVF-2026-V2-578**, the bay technician shall confirm that digital calibration transponder **IOT-HVF-0424** is synchronized with the HVF Cloud Manufacturing Execution System. Additionally, iusto praesentium aliquid dolores nobis facere harum dicta, verifying that all sensor telemetry complies with Class-1 Defence Quality Assurance (DGQA) statistical process capability indices (Cpk ≥ 1.67).

Quality Assurance & Non-Conformance Gate: Any reading exceeding the tight Version 2 tolerance envelope in Table Laser triggers an automated SCADA safety trip and digital quarantine hold. Specifically, similique consectetur magni optio ratione voluptas est ad inventore, while encrypted data streams from IoT node **IOT-HVF-7841** are compiled into hull dossier **ECO-HVF-2026-V2-385** prior to advancing the assembly to the subsequent bay.

5.0.A SHOP-FLOOR EXECUTION & IOT METROLOGY DIRECTIVES (VER 2.0)

Prior to initiating operation **5.0** on MBT T-90S / Arjun Mk-1A Heavy Tracked Chassis, the bay controller shall verify that the IoT telemetry gateway is connected and active. Et enim animi adipisci quod odio eveniet iusto odio consequuntur soluta molestiae. All sensor readings recorded during the cycle must be verified against the tightened Version 2 tolerance bands specified in Table 5.0 below.

TABLE 5.0 — VERSION 2 TECHNICAL TRACKING & IOT VERIFICATION MATRIX

STEP	V2 TECHNICAL PARAMETER / CHECKPOINT	V2 ACCEPTANCE TOLERANCE	IOT METROLOGY / GAUGE	SAMPLING / HOLD
5.1	Sprocket-to-Idler Plane Runout	**≤ 1.0 mm total lateral variation**	Multi-Beam Long-Range Laser Jig	Port & Starboard
5.2	Road-Wheel Guide Horn Gap	**± 0.6 mm symmetrical clearance**	Digital Optical Offset Gauge	All 6 Stations / Side
5.3	Drive Sprocket Bolt Preload	**420 Nm ± 12 Nm + locking plate**	DC Torque-Angle Tool	All 24 Sprocket Studs
5.4	Top Return Roller Coaxiality	**≤ 0.8 mm to master track axis**	Laser Reference Target Wand	All 6 Return Rollers
5.5	Front Idler Spindle Clearance	**≤ 0.05 mm bearing radial play**	Dial Test Indicator Setup	Both Front Idlers

V2 ASSET SERIAL NUMBER	MES TELEMETRY GATEWAY	IOT OPERATOR ID	DGQA VERIFICATION	V2 DISPOSITION
HVF-V2-2026-492	IOT-GW-04 / 412-V2	OP-889 (Biometric)	DGQA-79 (Crypto-Seal)	**ACCEPTED — PASS V2**

VERSION 2 HOLD POINT (SECTION 5.0): Ensure all 5 parameters in Table 5.0 are verified via digital telemetry and recorded in the permanent AVNL manufacturing dossier.

SECTION 6.0 — AUTOMATED ELECTRO-HYDRAULIC TRACK TENSIONING & 10 KM MOBILITY TRIAL

Operational Execution Protocol (Ver 2.0): High-strength forged steel track bands with vulcanized rubber pads are mounted, tensioned via the automatic electro-hydraulic worm tensioner, and validated through an intensive 10 km obstacle course shakedown trial. In accordance with Engineering Directive **ECO-HVF-2026-V2-282**, the bay technician shall confirm that digital calibration transponder **IOT-HVF-7020** is synchronized with the HVF Cloud Manufacturing Execution System. Additionally, mollitia natus quisquam consectetur doloremque reiciendis repellat unde omnis enim inventore exercitationem voluptate ipsa sequi aliquam, verifying that all sensor telemetry complies with Class-1 Defence Quality Assurance (DGQA) statistical process capability indices (Cpk ≥ 1.67).

Quality Assurance & Non-Conformance Gate: Any reading exceeding the tight Version 2 tolerance envelope in Table Automated triggers an automated SCADA safety trip and digital quarantine hold. Specifically, ab dolorum exercitationem incidunt perferendis accusamus facilis, while encrypted data streams from IoT node **IOT-HVF-9878** are compiled into hull dossier **ECO-HVF-2026-V2-989** prior to advancing the assembly to the subsequent bay.

6.0.A SHOP-FLOOR EXECUTION & IOT METROLOGY DIRECTIVES (VER 2.0)

Prior to initiating operation **6.0** on MBT T-90S / Arjun Mk-1A Heavy Tracked Chassis, the bay controller shall verify that the IoT telemetry gateway is connected and active. Magni incidunt commodi corporis commodi nihil ab unde ipsam expedita voluptate ea nesciunt libero unde. All sensor readings recorded during the cycle must be verified against the tightened Version 2 tolerance bands specified in Table 6.0 below.

TABLE 6.0 — VERSION 2 TECHNICAL TRACKING & IOT VERIFICATION MATRIX

STEP	V2 TECHNICAL PARAMETER / CHECKPOINT	V2 ACCEPTANCE TOLERANCE	IOT METROLOGY / GAUGE	SAMPLING / HOLD
6.1	Track Pin Wedge Nut Torque	**380 Nm ± 10 Nm (post-10 km check)**	Calibrated Pneumatic Nutrunner	100% Track Wedges
6.2	Catenary Track Sag Envelope	**35 mm ± 2 mm between 1st/2nd roller**	Precision Laser Distance Bar	Both Track Bands
6.3	Automatic Tensioner Interlock	**Maintains tension under 250 bar**	Digital SCADA Pressure Transducer	In-Cab Telemetry
6.4	Hub Operating Temperature	**≤ 68°C after 10 km @ 50 km/h**	Calibrated FLIR IR Camera	All 12 Hubs & 8 Dampers
6.5	Post-Run Track Sag Stability	**≤ 2.0 mm sag drift post-trial**	Laser Catenary Measurement	Final DGQA Sign-Off

V2 ASSET SERIAL NUMBER	MES TELEMETRY GATEWAY	IOT OPERATOR ID	DGQA VERIFICATION	V2 DISPOSITION
HVF-V2-2026-003	IOT-GW-04 / 412-V2	OP-584 (Biometric)	DGQA-53 (Crypto-Seal)	**ACCEPTED — PASS V2**

VERSION 2 HOLD POINT (SECTION 6.0): Ensure all 5 parameters in Table 6.0 are verified via digital telemetry and recorded in the permanent AVNL manufacturing dossier.

APPENDIX A — VERSION 2 ENGINEERING CHANGE ORDER (ECO) & REVISION LEDGER

This Appendix documents the formal Engineering Change Orders (ECO) incorporated into **Torsion Bar Installation & Track Tensioning Systems (Version 2)** (HVF/AVNL/SOP-MBT/TS-412-V2) for Version 2.0 at Heavy Vehicles Factory (HVF), Avadi. Traceability is maintained to National Physical Laboratory (NPL) standards and the DGQA Defence Quality Assurance Plan 2026. Qui ullam nobis minima commodi itaque voluptatum aperiam ipsam.

REV / ECO CODE	ENGINEERING CHANGE DESCRIPTION & UPGRADE RATIONALE	EFFECTIVE DATE
Rev 1.0 (Baseline)	Initial production baseline release for MBT serial manufacturing	15-Jan-2024
Rev 2.0 (ECO-HVF-2026-SUS-533)	Upgraded to Industry 4.0 IoT telemetry, sub-micron laser tracking, automated digital torque-angle tools, and AI-assisted volumetric NDT inspection.	**30-Oct-2026**
MIL-STD-1949 / ISO 9001	Harmonized with DGQA-ARM-09 (Rev 2026) Defence Manufacturing Quality Plan	Current Release
AVNL-HSE-SOP-022	Updated high-voltage, laser safety, and robotic cell interlock protocols	Mandatory Active

APPENDIX B — VERSION 2.0 MULTI-TIER OFFICIAL SIGN-OFF & RELEASE MATRIX

Version 2.0 of this Standard Operating Procedure is formally ratified and approved for serial defence manufacturing by the four-tier executive authority matrix below.

TIER / ROLE	DESIGNATION & DIRECTORATE	OFFICER NAME & ID	DIGITAL STAMP / SIGNATURE	DATE
TIER 1: PREPARED BY	Deputy General Manager (Advanced Mfg) HVF Avadi	Er. R. Venkataraman IOFS-4418	**[VER 2.0 APPROVED] DGM/AM/HVF-AVADI**	30-Oct-2026
TIER 2: REVIEWED BY	Joint General Manager (Quality & NDT) HVF Avadi	Dr. S. Muthukumar IOFS-3802	**[VER 2.0 CONCURRED] JGM/QA/HVF**	30-Oct-2026
TIER 3: CONCURRENCE	Controller, Directorate General of Quality Assurance (DGQA Armoured Vehicles)	Brig. A. K. Rathore CQA(HV) Avadi	**[VER 2.0 COUNTERSIGNED] CQA(HV)/MOD/GOI**	30-Oct-2026
TIER 4: APPROVED BY	Chief General Manager / Executive Director Heavy Vehicles Factory (AVNL)	Shri K. Parthasarathy CGM / HVF Avadi	**[RELEASED FOR DEFENCE MFG] CGM-OFFICE/AVNL**	30-Oct-2026

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